Assignment 8. Eigenvalues and eigenvectors Marks 20

Posted on 04.11.2026 @ 2:30 pm and due on 04.11.2026 @ 5:40 pm

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1. Prepare a library function for Power Iteration method to find eigenvalues and eigenvectors of a matrix when it is passed to it. **[8]**

2. Find the largest eigenvalue and its corresponding eigenvector of the following matrix using Power iteration method. **[6]**

$$\mathbf{A} = \begin{pmatrix} 4 & 2/3 & -4/3 & 4/3 \\ 2/3 & 4 & 0 & 0 \\ -4/3 & 0 & 6 & 2 \\ 4/3 & 0 & 2 & 6 \end{pmatrix}$$

3. Consider the 5×5 tridiagonal matrix given in the file `assign8.txt`. Using Power iteration method, verify that the first two largest eigenvalues and their corresponding eigenvectors of the matrix satisfy

$$\lambda_k = b + 2\sqrt{ac}\cos\left(\frac{k\pi}{n+1}\right) \quad \text{and} \quad v_k^i = 2\left(\sqrt{\frac{c}{a}}\right)^k \sin\left(\frac{ik\pi}{n+1}\right)$$

where $a = c = -1$, $b = 2$, $n = 5$, $k = 1, 2, \ldots 5$ and i is the i-th component of the k-th eigenvector. In case of any discrepancies, discuss its possible source(s). **[6]**